



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: GEOGRAPHY (SET A)

FULL MARKS: 80
TIME: 2 HOURS

Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.
Part I is compulsory. All questions from Part I are to be attempted.
A total of five questions are to be attempted from Part II.
The marks intended for questions are given in brackets [].
This paper consists of six printed pages.

PART I (30 marks)

(Attempt all questions from this Part.)

Question 1

Study the map extract G43S7 of the Survey of India and answer the following questions:

- (i) (a) Give the six figure grid reference of $\Delta 198$. [2]
(b) What is the drainage pattern in the grid square 2806?
- (ii) Mahesh worked in Delhi for many years and decided to settle in a relatively smaller town with good facilities after his retirement. Which settlement of the given map extract will he prefer to settle down and why? [2]
- (iii) (a) State the compass direction of Mahudi Moti from Jegol. [2]
(b) Calculate the area between Eastings 24 to 29 and Northings 03 to 08 in km.
- (iv) What type of rainfall is experienced in the given map area? Give evidence to support your answer. [2]
- (v) Rahul, who lives in Rampura (Panswala) went to Santarwada in a bullock cart to visit his uncle. If he moves at a speed of 5 kilometers per hour in a straight line, how much time will he take to reach his destination? [2]

Question 2

On the outline map of India provided: [10]

- (i) Draw and label Indian Standard Meridian.
- (ii) Mark and name the tributary of River Krishna.
- (iii) Mark and name the highest peak of India.
- (iv) Shade and label an area having alluvial soil south of Tropic of Cancer.
- (v) Mark and label Palk Strait.
- (vi) Mark and name the city located at the confluence of Rivers Ganga and Yamuna.
- (vii) Shade and label an area with sparse population in North Eastern India.
- (viii) Mark and name an offshore oil field.
- (ix) Draw arrows to show the direction of the winds that bring rainfall to Coromandel Coast in winter.
- (x) Mark and label Nathu La Pass.

Question 3

Choose the correct answers to the questions from the given options.
(Do not copy the question, write the correct answers only.)

[10]

- (i) Which of the following is NOT a region with scanty rainfall?
 - (a) Southern Punjab
 - (b) Western slopes of Western Ghats
 - (c) Western Rajasthan
 - (d) Parts of Ladakh

- (ii) Sandalwood is found in _____ forests.
 - (a) Tidal
 - (b) Tropical Deciduous
 - (c) Tropical Evergreen
 - (d) Mountain

- (iii) Name the state which is solely dependent on canals for irrigation.
 - (a) Maharashtra
 - (b) Odisha
 - (c) Karnataka
 - (d) Mizoram

- (iv) Red soil is ideal for dry farming because:
 - (a) It is formed in areas with poor rainfall.
 - (b) It is having more concentration of iron oxides.
 - (c) It is clayey in nature.
 - (d) It is poor in organic matters.

- (v) Which of the following is the best quality coal?
 - (a) Bituminous
 - (b) Lignite
 - (c) Anthracite
 - (d) Peat

- (vi) Which of the following is NOT a reason for the migration of Sugar Industries to South India?
 - (a) Excellent transport facilities in states of Maharashtra & Tamil Nadu.
 - (b) Bigger areas are managed by cooperative societies.
 - (c) Shorter crushing periods.
 - (d) Presence of fertile black soil.

- (vii) _____ is alloyed with zinc to form brass.
 - (a) Bauxite
 - (b) Iron ore
 - (c) Copper
 - (d) Manganese

(viii) Which of the following ports is on the west coast of India?

- (a) Mumbai
- (b) Visakhapatnam
- (c) Chennai
- (d) Paradip

(ix) _____ is the IT capital of India.

- (a) Mumbai
- (b) Hyderabad
- (c) Delhi
- (d) Bengaluru

(x) Which of the following is an advantage of wind energy?

- (a) It is a non renewable energy.
- (b) It pollutes the air.
- (c) It decreases the use of fossil fuels.
- (d) It increases the monthly electricity bills of farmers.

PART – II (50 marks)

(Attempt any five questions from this part)

Question 4

(i) *“Rain played spoilsport as the one-day cricket at Mohali (Punjab) match between India and South-Africa was halted here today. Keeping in view the inclement weather, the match was called off and both teams were given two points each.”*

The Tribune,

14th January 2022. [2]

(a) What caused this rainfall in Punjab during the winters?

(b) State the agricultural benefit of this rainfall?

(ii) Differentiate between Tropical Cyclones and Temperate Cyclones. [2]

(iii) Give the geographical reasons for each of the following: [3]

(a) Mangalore experiences an equable climate.

(b) Mumbai experiences vertical rays of the Sun twice a year whereas Delhi does not experience it even once.

(c) The Coromandel Coast remains dry during the monsoon season.

(iv) Study the climatic data given below and answer the following questions: [3]

MONTHS	J	F	M	A	M	J	J	A	S	O	N	D
Temp in °C	10.4	16.6	21.6	25.5	33.2	33.5	30.8	29.8	29.2	25.5	19.6	15.2
RF in CM	2.5	2.0	1.5	0.9	1.5	7.5	17.8	18.5	12.5	1.0	0.2	1.5

(a) Calculate the annual range of temperature.

(b) Calculate the amount of rainfall during the monsoon season.

(c) Suggest a probable name for this station.

Question 5

- (i) Differentiate between gully erosion and sheet erosion. [2]
- (ii) With reference to Laterite soil, answer the following: [2]
 - (a) How are they formed?
 - (b) Why are they useful for construction purposes?
- (iii) Name the soil: [3]
 - (a) Found in the Ganga- Brahmaputra delta.
 - (b) Which is also known as In-situ soil.
 - (c) Which suits the growth of tapioca and cashew nuts.
- (iv) Give geographical reasons for the following statements: [3]
 - (a) Red soil requires irrigation.
 - (b) Construction of dams controls soil erosion.
 - (c) Deltaic alluvium is more fertile than the coastal alluvium.

Question 6

- (i) *Development activities caused 41% decline in evergreen forests of central Western Ghats. Land use analysis of four neighbouring riverscapes in the central Western Ghats between 1973 and 2018 has revealed 41% decline in evergreen forests and 60% fragmentation of intact or contiguous forests, according to a new study, which also highlights the perils of degradation of riverine ecosystems – a primary cause for increasing water insecurity.*
thehindu.com, June 10, 2020
 - (a) State any two climatic conditions required for the Tropical Evergreen forest?
 - (b) Why are they difficult to exploit commercially? [2]
- (ii) State two importance of forests in protecting the environment. [2]
- (iii) Define Social Forestry. State its two objectives. [3]
- (iv) (a) Why should trees be planted around the cities having Iron & Steel plants?
(b) What is the difference between afforestation and reforestation? [3]

Question 7

- (i) Despite plenty of rain in India during the rainy season, why do we need irrigation? Give any two reasons. [2]
- (ii) State any two objectives of the Rainwater Harvesting. [2]
- (iii) Give geographical reasons for the following statements: [3]
 - (a) Canal irrigation makes the soil unproductive.
 - (b) Drip irrigation reduces loss of water through evaporation.
 - (c) Well irrigation is not preferred in the Peninsular Plateau region of India.
- (iv) With reference to tube well irrigation, answer the following questions: [3]
 - (a) Mention any two requirements for their construction.
 - (b) Why are they considered a more reliable source of irrigation when compared to well irrigation?

Question 8

- (i) Why is manganese a useful raw material for Iron & Steel Industries? Name one place in Odisha where iron ore is found. [2]
- (ii) With reference to Bhakra Nangal Project, answer the following: [2]
 - (a) On which river is it based?
 - (b) Name any two Indian states benefited by this project.
- (iii) Give two disadvantages of using Coal as a source of energy. Why is Anthracite coal considered best for domestic use? [3]
- (iv) State any two advantages of geothermal energy. How is India in an advantageous position to harness solar energy? [3]

Question 9

- (i) Why is there a need for rapid industrialization in India? Give two reasons. [2]
- (ii) From where does Rourkela Steel plant obtain the following : [2]
 - (a) Iron ore.
 - (b) Hydro Power supply.
- (iii) Give geographical reasons for the following statements: [3]
 - (a) Mumbai is known as the 'Cottonopolis of India'.
 - (b) Sugar mills should be located close to the sugarcane growing areas.
 - (c) Traditional materials are being replaced by petrochemical products.
- (v) With reference to silk industries, answer the following: [3]
 - (a) State any two problems faced by the silk industry.
 - (b) Name the types of silk found in Assam & Bihar.

Question 10

- (i) Name the terminal points of East-west and North-South corridors. [2]
- (ii) State any two economic benefits of the Golden Quadrilateral. [2]
- (iii) Mention any two advantages and one disadvantage of waterways. [3]
- (iv) Give the geographical reasons for the following statements: [3]
 - (a) Peninsular rivers are not ideal for navigation.
 - (b) Railway transport lacks flexibility.
 - (c) A well-developed transport network is important for industrial regions.

Question 11

- (i) Differentiate between khadar soil and bhangar soil. [2]
- (ii) State geographical conditions necessary for the growth of Littoral forest. [2]
- (iii) Which modern means of irrigation is also called overhead irrigation? State one advantage and one disadvantage of such an irrigation system. [3]
- (iv) Give geographical reasons for the following statements: [3]
 - (a) Railways are not common in north east India
 - (b) A large Indian population does not use air transport.
 - (c) Northern plains have a dense network of roads.

Question 12

- (i) State the significance of the Himalayas in the climate of India. [2]**
- (ii) Name the following: [2]**
 - (a) Oldest oil refinery in India**
 - (b) Ore from which aluminium is extracted.**
- (iii) State any three problems faced by the Iron & Steel Industry in India. [3]**
- (iv) Give geographical reasons for the following statements: [3]**
 - (a) Shimla is cooler than Delhi even during the summers.**
 - (b) Natural gas is a fossil fuel.**
 - (c) Petrochemical industries are located close to the oil refineries.**